ENGINEERING INNOVATION CHALLENGE 2023

ARIA™

Team code: J-27

School: Raffles Institution

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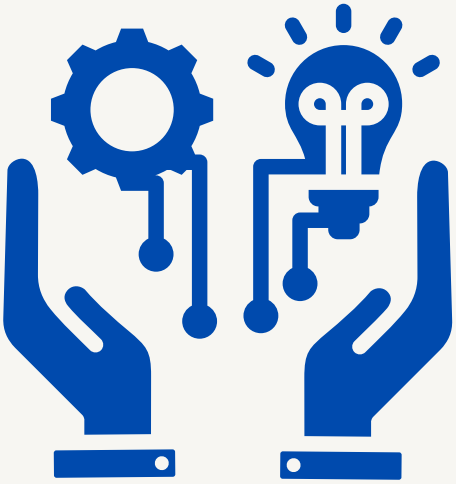
ABSTRACT •

Our group has chosen the topic of health innovation, and more specifically, we will be looking into the area of mental health which has been under the spotlight in recent years. Our group ultimately aims to utilise generative AI to improve the mental health of the elderly, since they are particularly more vulnerable to conditions such as loneliness and depression.

Our device will be placed in the home of elderly users, and will learn to have a simple conversation with them. The device will analyse these conversations and understand how the elderly is feeling and their general state of mind. The device will then report these findings to an app on a daily basis for users/caregivers to understand the well-being of users. Moreover, it will also provide tips to improve well-being and have a positive outlook on things when conversing with the elderly.



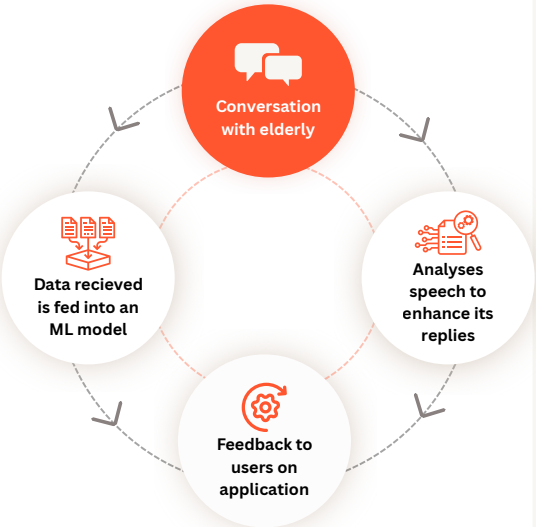
ARIA™

Background and purpose •

As per the American Psychological association, 84% of psychologists who treat anxiety disorders say there's been an increase in demand for treatment since the start of the pandemic. Elderly people who perceive themselves to be lonely can expect to live three to five years less on average, compared to peers who perceive themselves as "not lonely" (Duke-NUS, 2021). It is generally accepted that individuals who have strong social connections to family, friends and community, age more healthily than those with weak or no social connections (Ward et al., 2020). Thus, we will be looking at how we can effectively use Artificial intelligence (AI) to improve one's mental health which is a pressing issue in our society. Our target audience will be those who are more vulnerable to mental health-related conditions such as the elderly. AI in healthcare is a novel and rapidly growing solution, and our project aims to improve the mental well-being of the elderly, so that they can stay positive and through the collection of data, we can monitor progress and track improvements as well. Moreover, using AI also decreases the burden placed on caregivers who need to be constantly looking after the elderly. When it comes to resilient city living, our group believes that the mental well-being of the population has a direct relation to a society's ability to be able to rebound from challenges such as COVID-19. Hence, resilient cities are those that choose to invest in the mental health of their people, especially the elderly, in a post-pandemic world by taking advantage of AI.

• Design proposal

Our design proposal is to develop a small physical device placed in the homes of elderly individuals, which would communicate with them to gain a better understanding of their life and mood. This device would be connected to power and the internet, enabling its functionality. Equipped with a lidar sensor, the device would be capable of detecting when the elderly person is in its vicinity. It would then attempt to initiate a conversation with a friendly greeting, patiently awaiting a response. Through human-like interaction, the device would strive to provide companionship to the elderly individual. Subsequently, the audio recording would be broken into useful components and analysed to categorise it into a type of human emotion. Our device thus facilitates connectivity between family members, enabling them to stay connected despite their hectic schedules. The three main goals of our project are as follows; Firstly, our device should be able to have a simple conversation with the elderly. Secondly, our device needs to analyse the inputs from the elderly and report it back to a functioning app. The app must infer the data and display it for users such as the elderly's caretakers to understand the state of the elderly's mental health. Ultimately, we hope our AI device can help reduce the likelihood of the elderly developing mental health illnesses like depression and dementia by keeping them constantly engaged. It can remind them to take their medicines on time and remind them of upcoming events like birthdays, just having a conversation about how they are doing could lessen the loneliness faced by these old people and keep them mentally sharp.



Innovation tech stack • Machine Learning Models

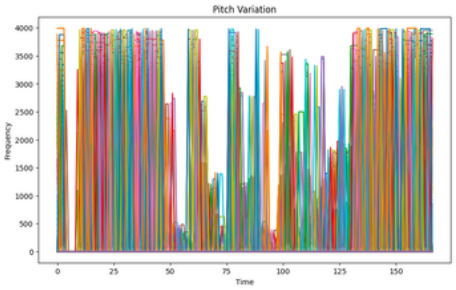
In the context of human-computer interaction, emotion recognition can help play a pivotal role in understanding and producing nuanced and meaningful conversation. To accurately categorise audio samples into various emotions, we extract numerous features from an audio sample. We deployed a multi-faceted approach to understanding sound. We aimed to understand sound from a psychological, biological, and mathematical perspective. We started by understanding the process of speaking and listening.

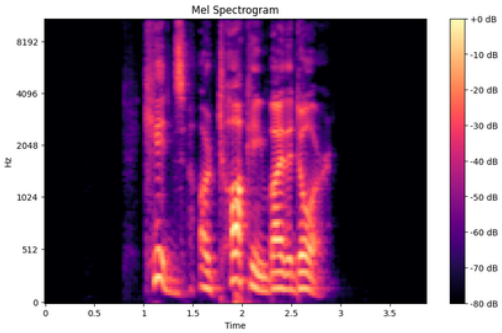
Data Collection and Preparation •

We found a dataset of an equal number of male and female voice actors speaking in various emotions with varying intensity. Here is a link to the dataset. Our dataset was then preprocessed into 8 emotions (neutral, calm, happy, angry, sad, disgusted, fearful, surprised) all of which there are an exactly equal number of, preventing overfittingbt.

Feature Extraction •

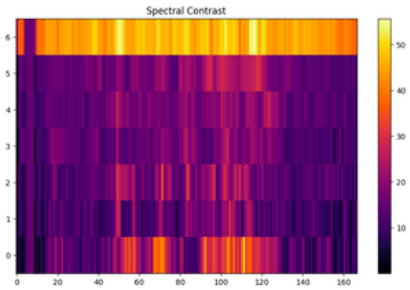
Our bodily response to stress may trigger physiological changes in our body, such as increased heart rate, muscle tension, and respiratory rate. Intense emotions can cause increased muscle activity in the vocal cords, leading to variations in vocal cord tension in the laryngeal muscles, leading to variation in pitch.

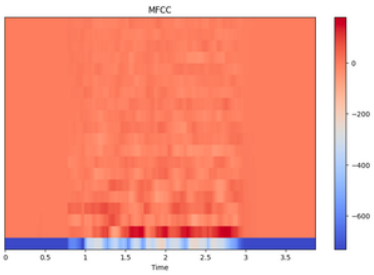


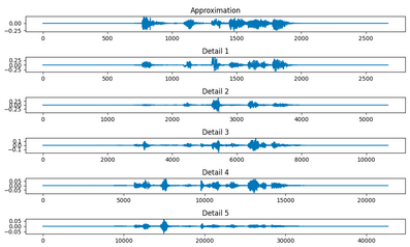


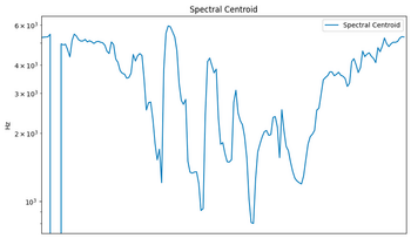
Humans can tell certain frequencies apart better than others as a result of the biology of our ear's cochlea. This means certain frequencies are more important in conveying and understanding emotion than others. Mathematically, this is plotted by the Mel Scale. By applying that on top of a spectrogram of the audio, we can extract highlight important data and suppress noise.

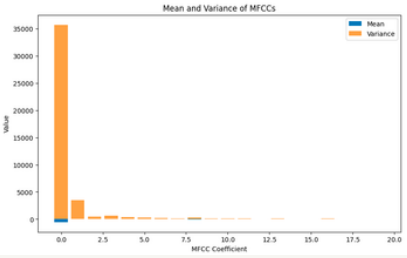
Below are the other 6 extracted features, each highlighting a different characteristic of speech that can help understand the feelings of the speaker. These features were extracted with the help of Fourier Transforms, Cepstrum Analysis, and many more mathematical operations.

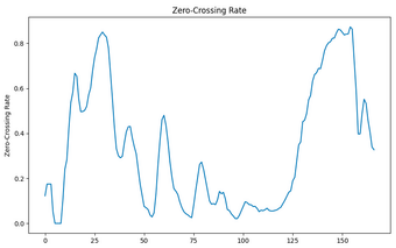












Feature Scaling •

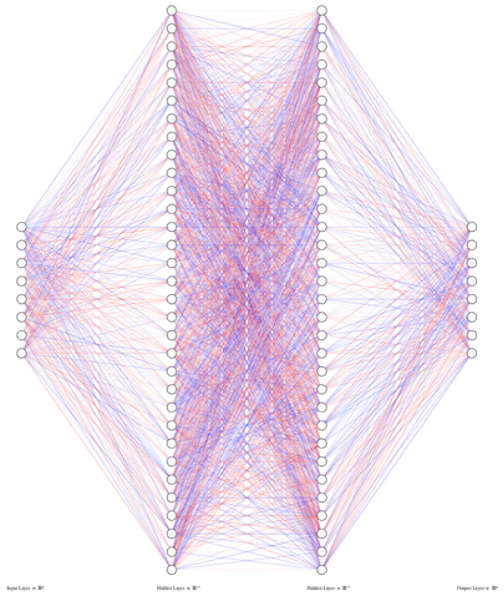
Not all features are created equal, so we used Z-score normalisation for the process of feature scaling allowing all features to be normalised such that no specific metric has dominance over others due to it having a larger magnitude. This helps prevent any biases in the model and produces a more fair and accurate result. In addition, this speeds up the rate of convergence of our model (decreasing the training time required to optimize it)

Model Selection •

Our model is working on a multi-class classification problem (happy, sad, fearful, etc.). The most commonly used model for this type of problem in the industry is a Support Vector Machine (SVM). We used 8 SVM's, one for each extracted feature so as to understand what characteristics of each feature help identify its emotion. For our meta model, we are using a neural network to capture the nuanced relationships between the individual models' predictions and emotions.

Meta-Model and Results Aggregation •

A meta model is used, taking the outputs of each individual SVM model as an input. By harnessing a convolutional neural network (CNN), we are able to weigh each feature's importance in determining emotion as a measure of its "correctness" across a large dataset. Our neural network uses two hidden layers with 32 neurons each with 2 MaxPooling layers between them (Not Shown) used to downsize the data and speed up the training process. The input layer has 8 neurons because this meta CNN model takes the 8 outputs of each SVM model as an input and 8 neurons in the output layer as the model must assign the audio recording into one of 8 emotions that it has been trained on. Then, our trained meta model produces the final output sent to the app as a positive number if a positive emotion has been classified and negative if a negative emotion has been classified with the model's percentage confidence as a measure of the intensity their mood displayed as a percentage change up or down in the app.



Graph of Neural Network Architecture

Product • Prototype Modelling

The Raspberry Pi, at its core, is a miniature computer but with profound implications. Here's an elucidative breakdown of how we incorporated and utilized the Raspberry Pi in our project:

Hardware integration •

Lidar Sensor: We integrated a Lidar sensor with the Raspberry Pi through its GPIO (General-Purpose Input/Output) pins. This sensor acts as our primary trigger, notifying the Raspberry Pi when the elderly individual is nearby. We equipped the Raspberry Pi with a microphone and speaker setup. The microphone captures the auditory inputs from the user, while the speaker facilitates the AI-driven human-like interactions. The Raspberry Pi is connected to the internet via its onboard Wi-Fi module. This is paramount for real-time data processing and transmission to the dedicated application.

Software framework •

Operating System: We utilized Raspberry Pi OS, which is a Debian-based operating system. This offers stability, versatility, and compatibility with a plethora of open-source libraries and tools for us to work with.

AI Interface: Our emotion recognition system, built on the open-source Mycroft AI framework, is hosted on the Raspberry Pi. This system processes the audio, discerns emotions, and drives the conversational interactions.

Data Transmission: Python scripts running on the Raspberry Pi govern the processing of audio data and its transmission to our backend server using APIs to send data to both the machine learning and reply generating functions . Here, the raw data is further processed, and insights are transmitted to the dedicated application for caregivers or family members to review..

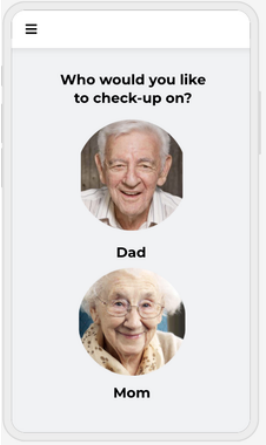
Connection Workflow •

Once the lidar sensor detects the presence of an individual, it sends a signal to the Raspberry Pi. The Raspberry Pi, upon receiving this signal, initializes the Mycroft AI interface and sends out a friendly greeting via the connected speaker. As the elderly individual responds, the microphone captures this auditory input, which is then processed by our emotion recognition system on the Raspberry Pi.

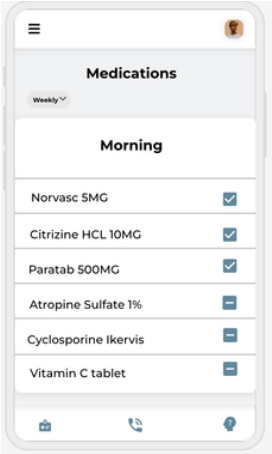
Engineering Implications •

We chose the Raspberry Pi because it's practical and rooted in real-world engineering. Its blend of hardware and software showcases the depth of our engineering work. This choice ensures our solution is both solid in concept and reliable in practice.

Reporting app for caregivers •

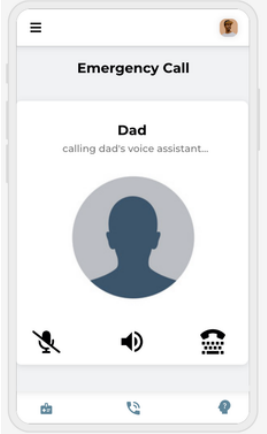


Profile page



Medicines tab

Dashboard



Call function

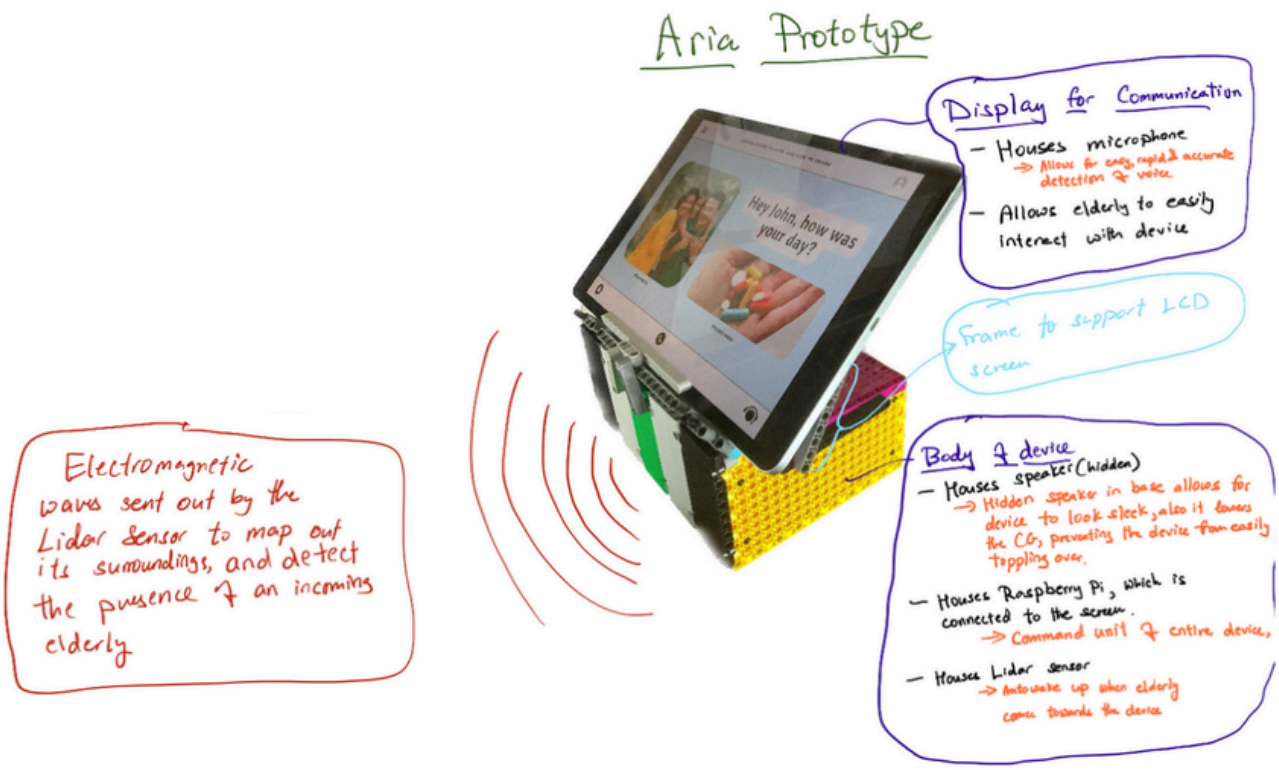
App description •

Our app bridges the gap between the elderly and their caretakers. As can be seen in the model above, caretakers can select which profile they'd like to enter, and view their dependents information from there. The medications tab allows caretakers to ensure that the right medications are being taken at the right times, information which is gathered through conversations with the elderly. The overall dashboard allows caretakers to get a quick-view on how their dependents are feeling, with information such as their mood and wellbeing. The last tab allows caretakers to call their dependents directly through the Aria device, allowing them to easily contact them in emergency situations. Overall, the app is designed with the end-user in mind, with user interference of the app being simple and intuitive, providing all the necessary features required to allow caretakers to monitor the health of their dependents, anytime, anywhere.

Evaluation •

Overall, we hope our project emphasises the importance of using Artificial intelligence to improve the mental health of our elderly. We strongly believe that such solutions will become an integral part of the elderly's daily lifestyle, and there are many more features which can be added to our project in the future. One such example, is the ability to detect sudden falls using a sensor system inbuilt into Aria, which will quickly send an alert to caretakers. As a nation which aims to progress together as one, it is essential that we do not leave behind anyone, especially the elderly who are more likely to suffer from loneliness and other health conditions.

ARIA Device Prototype •



APPENDIX •

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